

1. Grade Checker

Take a score as input and print the grade based on the following:

90+ : "A"

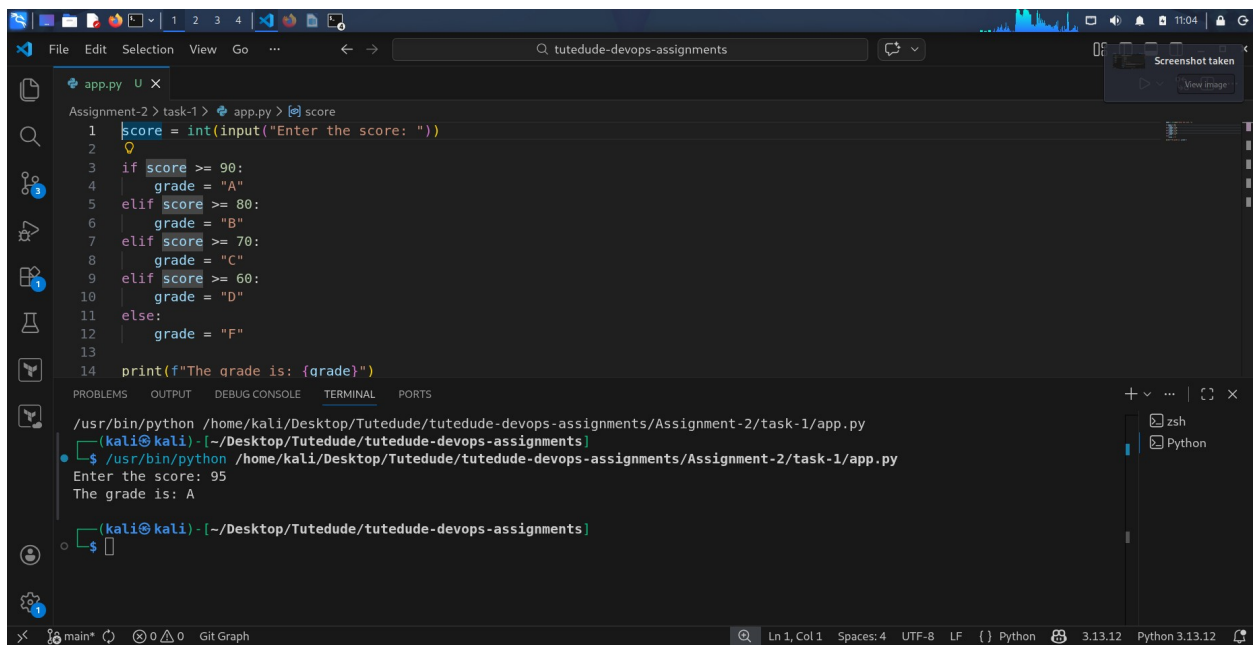
80-89 : "B"

70-79 : "C"

60-69 : "D"

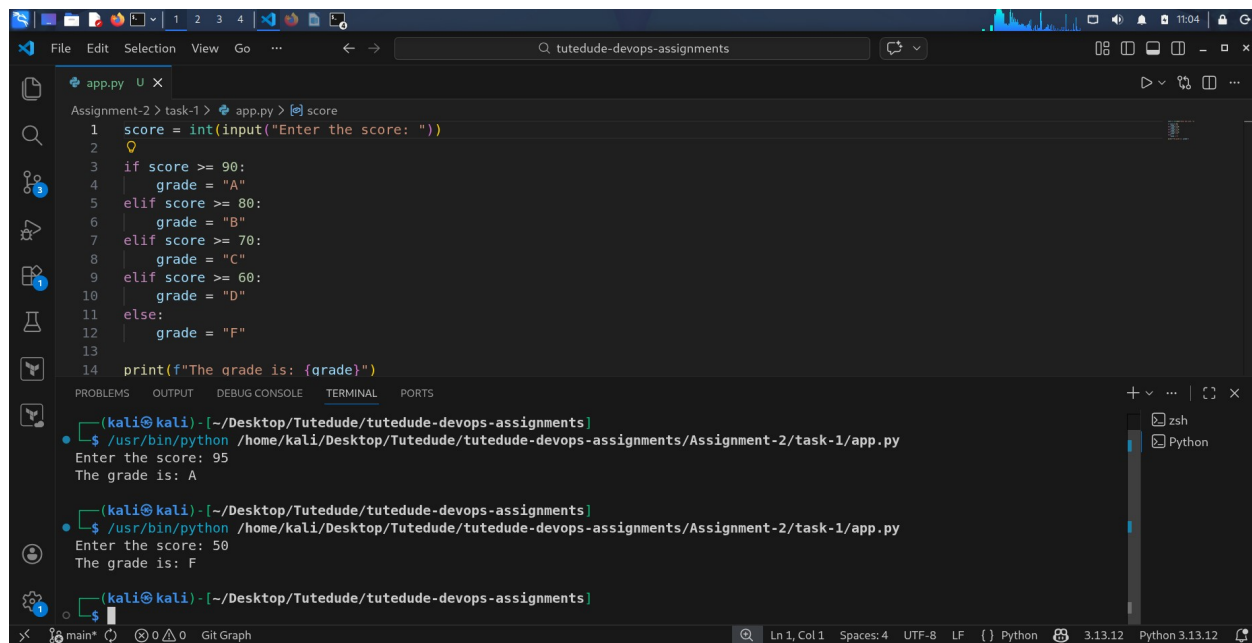
Below 60 : "F"

here we used a basic if else statement to carry out marks and all.



```
app.py
Assignment-2 > task-1 > app.py > score
1 score = int(input("Enter the score: "))
2
3 if score >= 90:
4     grade = "A"
5 elif score >= 80:
6     grade = "B"
7 elif score >= 70:
8     grade = "C"
9 elif score >= 60:
10    grade = "D"
11 else:
12    grade = "F"
13
14 print(f"The grade is: {grade}")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
/usr/bin/python /home/kali/Desktop/Tutodude/tutodude-devops-assignments/Assignment-2/task-1/app.py
(kali@kali): ~/Desktop/Tutodude/tutodude-devops-assignments
$ /usr/bin/python /home/kali/Desktop/Tutodude/tutodude-devops-assignments/Assignment-2/task-1/app.py
Enter the score: 95
The grade is: A
(kali@kali): ~/Desktop/Tutodude/tutodude-devops-assignments
$
```



The image shows a Visual Studio Code editor window with a Python file named `app.py` open. The code is a simple script that takes a score as input and prints the corresponding grade. The terminal at the bottom shows the script being executed twice: first with a score of 95, resulting in a grade of 'A', and then with a score of 50, resulting in a grade of 'F'.

```
Assignment-2 > task-1 > app.py > score
1 score = int(input("Enter the score: "))
2
3 if score >= 90:
4     grade = "A"
5 elif score >= 80:
6     grade = "B"
7 elif score >= 70:
8     grade = "C"
9 elif score >= 60:
10    grade = "D"
11 else:
12    grade = "F"
13
14 print(f"The grade is: {grade}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- (kali@kali) - [~/Desktop/Tutodude/tutodude-devops-assignments]
\$ /usr/bin/python /home/kali/Desktop/Tutodude/tutodude-devops-assignments/Assignment-2/task-1/app.py
Enter the score: 95
The grade is: A
- (kali@kali) - [~/Desktop/Tutodude/tutodude-devops-assignments]
\$ /usr/bin/python /home/kali/Desktop/Tutodude/tutodude-devops-assignments/Assignment-2/task-1/app.py
Enter the score: 50
The grade is: F
- (kali@kali) - [~/Desktop/Tutodude/tutodude-devops-assignments]
\$

Ln 1, Col 1 Spaces: 4 UTF-8 LF Python 3.13.12 Python 3.13.12

2 Student Grades

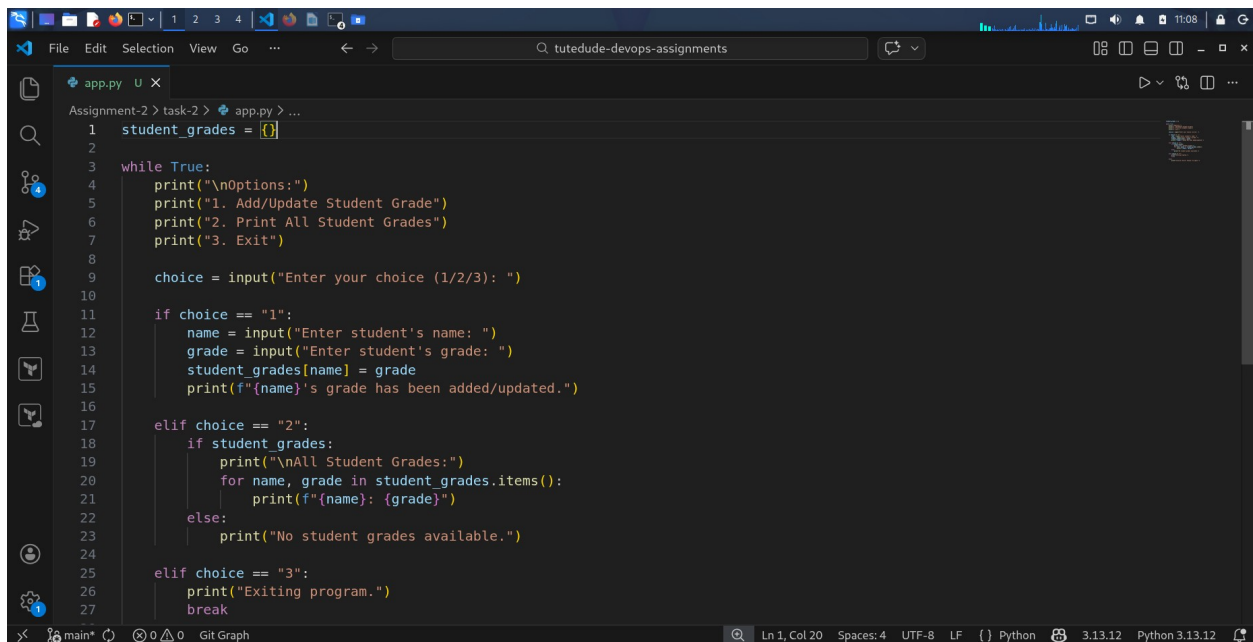
Create a dictionary where the keys are student names and the values are their grades.

Allow the user to:

Add a new student and grade.

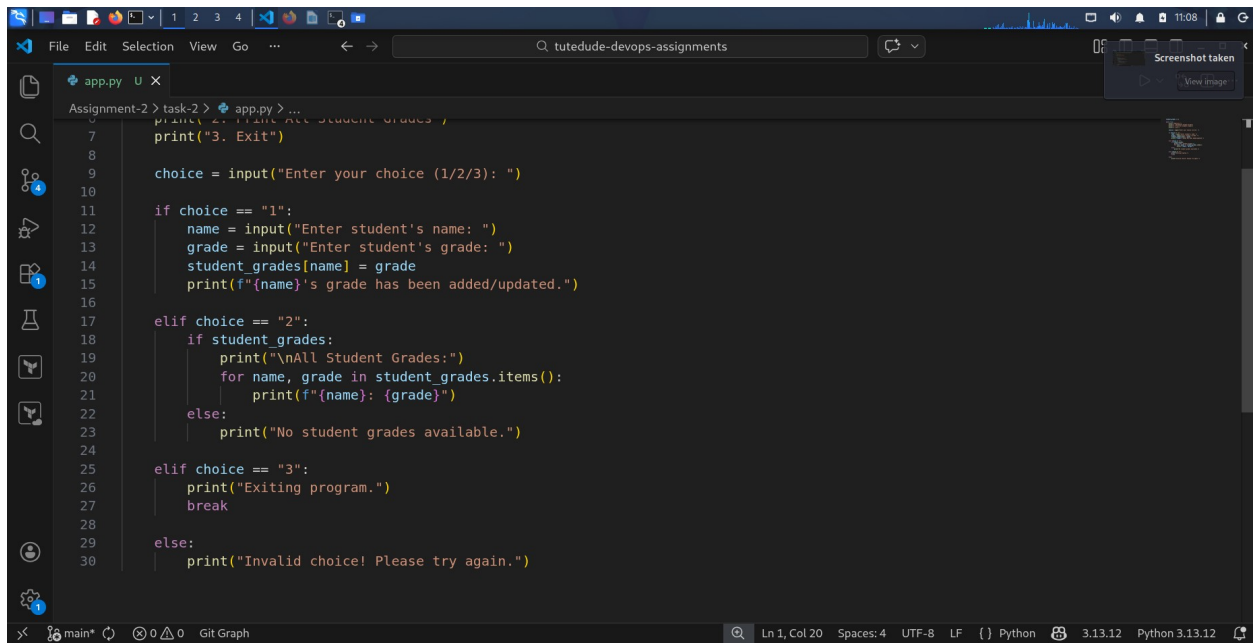
Update an existing student's grade.

Print all student grades.



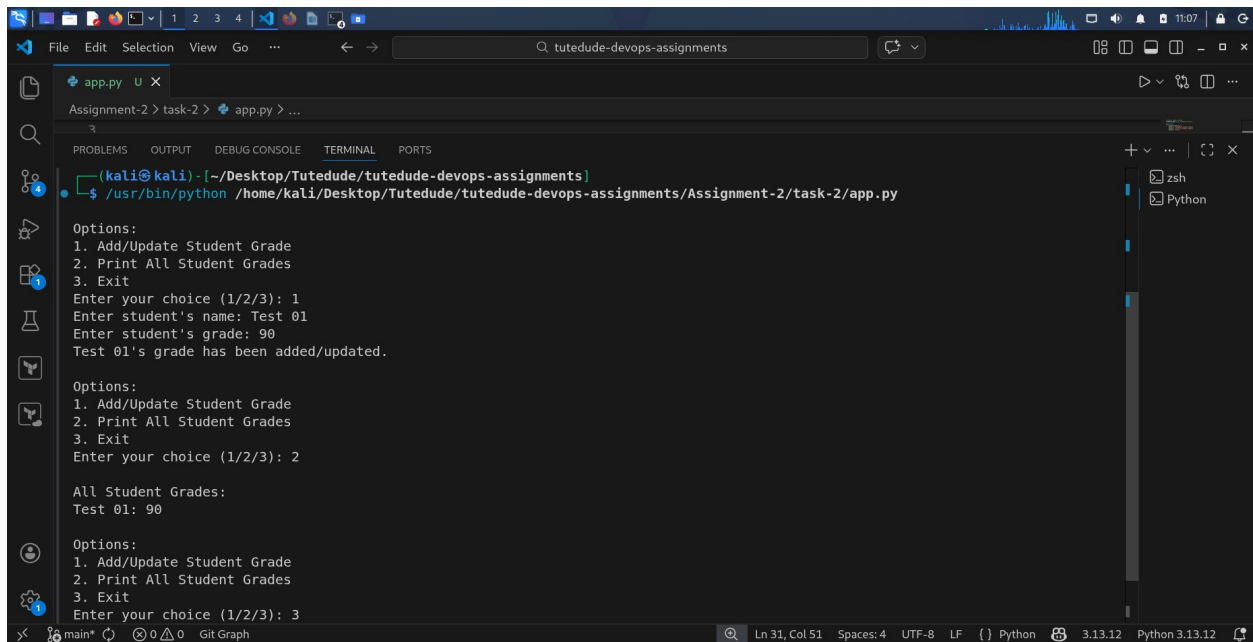
The screenshot shows a code editor window with a dark theme. The file explorer on the left shows a file named 'app.py'. The code in the editor is a Python script for managing student grades. It starts with an empty dictionary 'student_grades'. A 'while True' loop provides a menu with three options: '1. Add/Update Student Grade', '2. Print All Student Grades', and '3. Exit'. The user is prompted to enter a choice. If choice '1' is selected, the user is prompted for a student name and grade, which are then added to the dictionary. If choice '2' is selected, all student grades are printed. If choice '3' is selected, the program exits.

```
1 student_grades = {}
2
3 while True:
4     print("\nOptions:")
5     print("1. Add/Update Student Grade")
6     print("2. Print All Student Grades")
7     print("3. Exit")
8
9     choice = input("Enter your choice (1/2/3): ")
10
11     if choice == "1":
12         name = input("Enter student's name: ")
13         grade = input("Enter student's grade: ")
14         student_grades[name] = grade
15         print(f"{name}'s grade has been added/updated.")
16
17     elif choice == "2":
18         if student_grades:
19             print("\nAll Student Grades:")
20             for name, grade in student_grades.items():
21                 print(f"{name}: {grade}")
22         else:
23             print("No student grades available.")
24
25     elif choice == "3":
26         print("Exiting program.")
27         break
```



```
Assignment-2 > task-2 > app.py > ...
print(2. Print All Student Grades ,
print("3. Exit")

7
8
9     choice = input("Enter your choice (1/2/3): ")
10
11     if choice == "1":
12         name = input("Enter student's name: ")
13         grade = input("Enter student's grade: ")
14         student_grades[name] = grade
15         print(f"{name}'s grade has been added/updated.")
16
17     elif choice == "2":
18         if student_grades:
19             print("\nAll Student Grades:")
20             for name, grade in student_grades.items():
21                 print(f"{name}: {grade}")
22         else:
23             print("No student grades available.")
24
25     elif choice == "3":
26         print("Exiting program.")
27         break
28
29     else:
30         print("Invalid choice! Please try again.")
```



```
Assignment-2 > task-2 > app.py > ...

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

(kali@kali) - [~/Desktop/Tutodude/tutodude-devops-assignments]
$ /usr/bin/python /home/kali/Desktop/Tutodude/tutodude-devops-assignments/Assignment-2/task-2/app.py

Options:
1. Add/Update Student Grade
2. Print All Student Grades
3. Exit
Enter your choice (1/2/3): 1
Enter student's name: Test 01
Enter student's grade: 90
Test 01's grade has been added/updated.

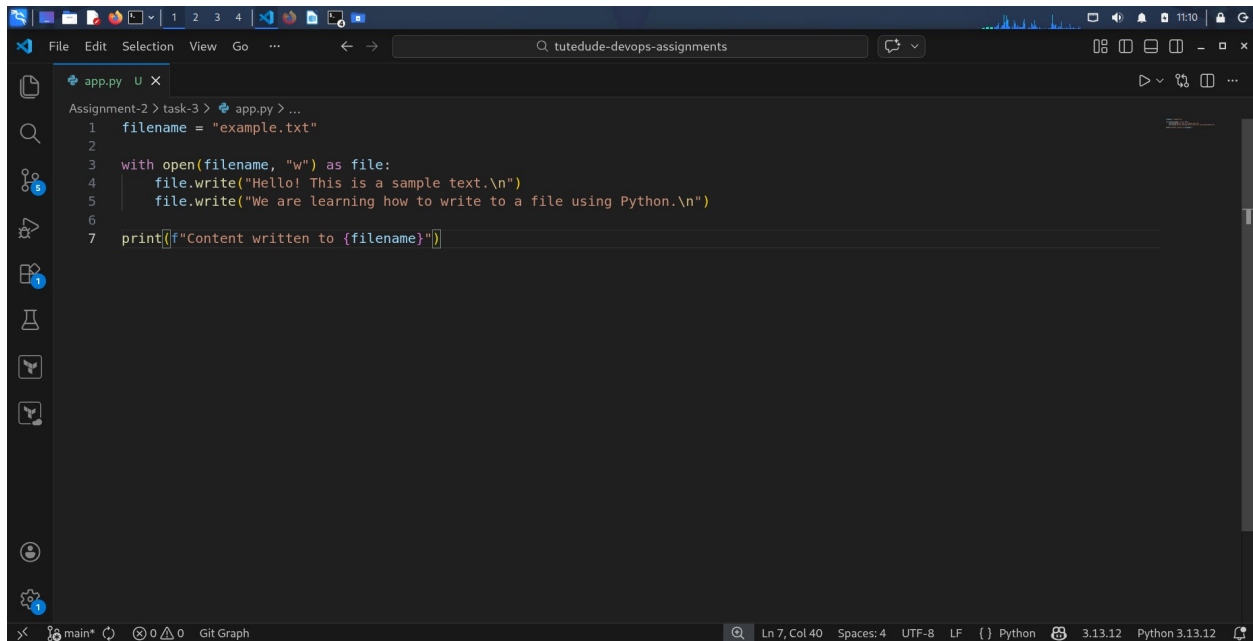
Options:
1. Add/Update Student Grade
2. Print All Student Grades
3. Exit
Enter your choice (1/2/3): 2

All Student Grades:
Test 01: 90

Options:
1. Add/Update Student Grade
2. Print All Student Grades
3. Exit
Enter your choice (1/2/3): 3
```

3. Write to a File

Write a program to create a text file and write some content to it.



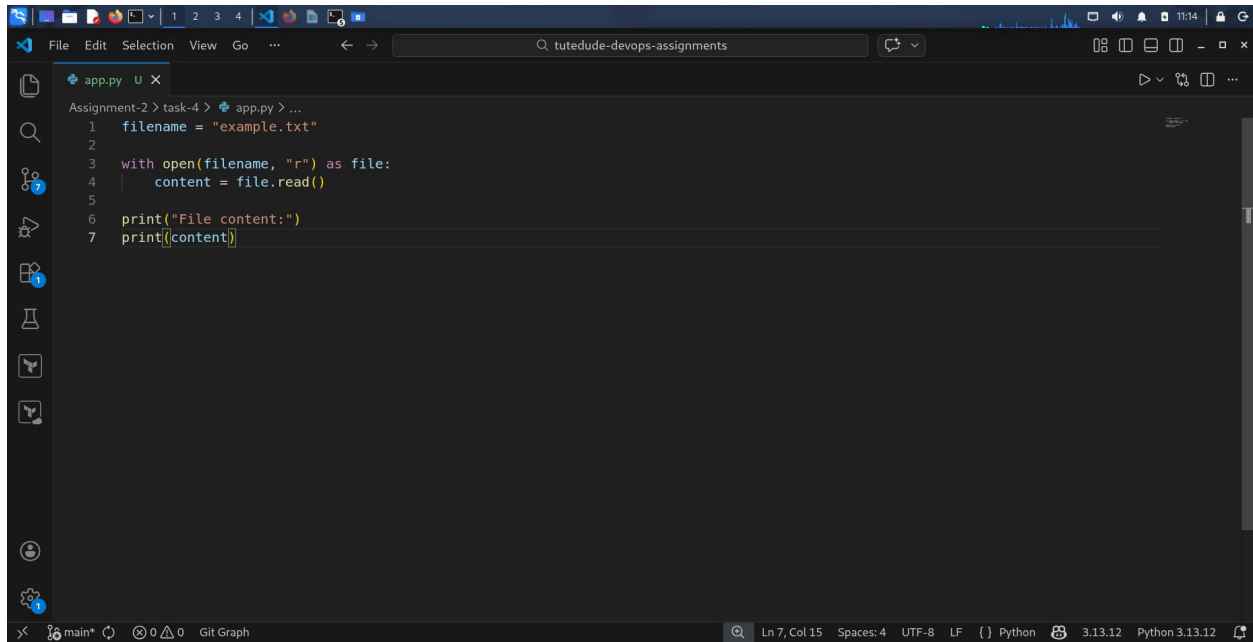
The image shows a screenshot of the Visual Studio Code (VS Code) editor interface. The editor is open to a file named `app.py`. The code in the editor is a Python script that creates a text file named `example.txt` and writes two lines of text to it. The code is as follows:

```
1 filename = "example.txt"
2
3 with open(filename, "w") as file:
4     file.write("Hello! This is a sample text.\n")
5     file.write("We are learning how to write to a file using Python.\n")
6
7 print(f"Content written to {filename}")
```

The VS Code interface includes a sidebar on the left with icons for Explorer, Search, Source Control, and Run and Debug. The top of the editor shows the file explorer with `app.py` selected. The bottom status bar indicates the current file is `main*`, the cursor is at line 7, column 40, and the file is encoded in UTF-8 with LF line endings. The Python version is 3.13.12.

4. Read from a File

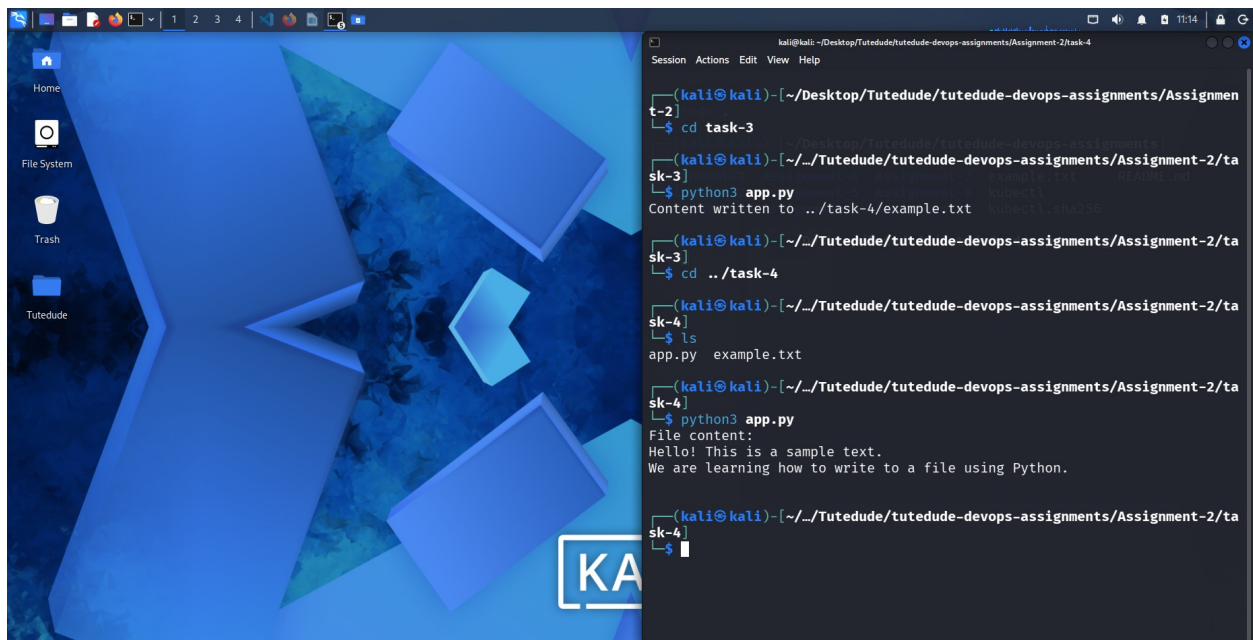
We used open in read mode and file.read to read and print to display.



The screenshot shows a code editor with a dark theme. The file 'app.py' is open, showing the following Python code:

```
1 filename = "example.txt"
2
3 with open(filename, "r") as file:
4     content = file.read()
5
6 print("File content:")
7 print(content)
```

The editor's status bar at the bottom indicates the file is on the 'main' branch, with 0 changes, and is using Python 3.13.12.



The screenshot shows a terminal window with a Kali Linux desktop background. The terminal session is as follows:

```
kali@kali: ~/Desktop/Tutodude/tutodude-devops-assignments/Assignment-2/task-4
$ cd task-3
$ python3 app.py
Content written to ../task-4/example.txt
$ cd ../task-4
$ ls
app.py  example.txt
$ python3 app.py
File content:
Hello! This is a sample text.
We are learning how to write to a file using Python.
```